2.54 Forces: Connected Particles

Worked Solutions

1 a)

On A:

$$\begin{aligned} \text{R}(\uparrow): \quad R - 5 \times 9.8 = 0 &\Rightarrow R = 49 \\ F &= 0.25 \times 49 = 12.25 \\ \text{R}(\rightarrow): \quad T - 12.25 &= 5a \end{aligned}$$

On B:

$$\begin{aligned} \text{R}(\downarrow): \quad 3 \times 9.8 - T &= 3a \\ -12.25 + 29.4 = 8a &\Rightarrow a = \frac{343}{160} = 2.14 \text{ ms}^{-2} \text{ (3 SF)} \end{aligned}$$

$a = 2.1 \text{ ms}^{-2}$ (2 SF – AQA)

1 b)

$$T - 12.25 = 5 \times \frac{343}{160} \Rightarrow T = \frac{735}{32} = 23.0 \text{ (3 SF)}$$

$T = 23$ N (2 SF – AQA)

2 a)

On B:

$$\text{R}(\downarrow): \quad 5 \times 9.8 - T = 5 \times 0.02 \Rightarrow T = 48.9 \text{ N}$$

$T = 50$ N (2 SF – AQA)

2 b)

On A:

$$\begin{aligned} \text{R}(\rightarrow): \quad 48.9 - F = 12 \times 0.02 &\Rightarrow F = 48.66 \\ \text{R}(\uparrow): \quad R - 12 \times 9.8 = 0 &\Rightarrow R = 117.6 \\ 48.66 = 117.6\mu &\Rightarrow \mu = \frac{811}{1960} = 0.414\ldots \end{aligned}$$

$\mu = 0.41$ (2 SF – AQA)

2 c)

$$0.5 = 0 + \frac{1}{2} \times 0.02t^2 \Rightarrow t = 5\sqrt{2} = 7.07 \text{ seconds}$$

7.1 s (2 SF – AQA)

2 d)

0.5 m

3 a)

On A:

$$\begin{aligned} \text{R}(\uparrow): \quad R - 4 \times 9.8 = 0 &\Rightarrow R = 39.2 \\ F &= 0.4 \times 39.2 = 15.68 \\ \text{R}(\rightarrow): \quad T - 15.68 &= 4a \end{aligned}$$

On B:

$$\begin{aligned} \text{R}(\downarrow): \quad 5 \times 9.8 - T &= 5a \\ -15.68 + 49 = 9a &\Rightarrow a = \frac{833}{225} = 3.70 \text{ ms}^{-2} \end{aligned}$$

$a = 3.7$ ms^{-2} (2 SF – AQA)

3 b)

$$T - 15.68 = 4 \times \frac{833}{225} \Rightarrow T = \frac{1372}{45} = 30.4888\ldots$$

$T = 30.5$ N (3 SF)

$T = 30$ N (2 SF – AQA)

3 c)

$$s = 0 + \frac{1}{2} \times \frac{833}{225} \times 1.05^2 \Rightarrow s = 2.04085$$

$s = 2.04$ metres

$s = 2.0$ metres (2 SF – AQA)

4 a)

On A:

$$\begin{aligned} \text{R}(\uparrow): \quad R - 2g = 0 &\Rightarrow R = 2g \\ F &= 0.3 \times 2g = 0.6g \\ \text{R}(\rightarrow): \quad T - 0.6g &= 2a \end{aligned}$$

On B:

$$\begin{aligned} \text{R}(\downarrow): \quad mg - T &= ma \\ mg - 0.6g = (2+m)a &\Rightarrow a = \frac{mg - 0.6g}{2+m} \text{ ms}^{-2} \end{aligned}$$

4 b)

$$\begin{aligned} \frac{mg - 0.6g}{2+m} &\geq 0 \\ mg - 0.6g &\geq 0 \\ m &\geq 0.6 \end{aligned}$$

Minimum value of m is 0.6

5)

On B:

$$\mathrm{R}(\uparrow): \quad T - 9.8g = ma \Rightarrow T = 9.8m + ma$$

On A:

$$\begin{aligned}
\mathrm{R}(\uparrow): \quad R - 3m \times 9.8 = 0 &\Rightarrow R = 29.4m \\
F = \mu R &= 29.4\mu m \\
\mathrm{R}(\leftarrow): \quad P - 29.4\mu m - T &= 3ma \\
P - 29.4\mu m - 9.8m - ma &= 3ma \\
P &= 29.4\mu m + 4ma + 9.8m \\
P &= (29.4\mu + 4a + 9.8)m \text{ as req}
\end{aligned}$$

6 a)

On A:

$$\mathrm{R}(\uparrow): \quad R + 10\sin 30^\circ - 2 \times 9.8 = 0 \Rightarrow R = 14.6$$

$$F = 0.2 \times 14.6 = 2.92 \text{ N}$$

$F = 3.0$ N (2 SF – AQA)

6 b)

On B:

$$\mathrm{R}(\uparrow): \quad T - 0.5 \times 9.8 = 0.5a$$

On A:

$$\mathrm{R}(\leftarrow): \quad 10\cos 30^\circ - T - 2.92 = 2a$$

$$-4.9 + 5\sqrt{3} - 2.92 = 2.5a \Rightarrow a = 0.336 \text{ ms}^{-2}$$

$a = 0.34$ ms^{-2} (2 SF – AQA)

6 c)

$$T - 4.9 = 0.5 \times 0.336 \Rightarrow T = 5.07 \text{ N (3 SF)}$$

$T = 5.1$ N (2 SF – AQA)

6 d)

On B:

$$\mathrm{R}(\uparrow): \quad T - m \times 9.8 = ma$$

On A:

$$\begin{aligned}
\mathrm{R}(\leftarrow): \quad 10\cos 30^\circ - T - 2.92 &= 2a \\
-9.8m + 5\sqrt{3} - 2.92 &= (2+m)a \\
\Rightarrow a &= \frac{-9.8m + 5\sqrt{3} - 2.92}{2+m} \\
-9.8m + 5\sqrt{3} - 2.92 &> 0 \\
5\sqrt{3} - 2.92 &> 9.8m \\
m &< 0.5857402079
\end{aligned}$$

m is less than 0.586 kg (3 SF)

m is less than 0.59 kg (2 SF – AQA)

7 a)

On A:

$$\mathrm{R}(\downarrow): \quad 2 \times 9.8 - T_A = 2a$$

On B:

$$\mathrm{R}(\uparrow): \quad R - 8 \times 9.8 = 0 \Rightarrow R = 78.4$$
$$F = 78.4 \times 0.1 = 7.84$$
$$\mathrm{R}(\leftarrow): \quad T_A - 7.84 - T_B = 8a$$

On C:

$$\mathrm{R}(\uparrow): \quad T_B - 0.5 \times 9.8 = 0.5a$$

Solve simultaneously:

$$19.6 = T_A + 2a$$
$$-7.84 = -T_A + T_B + 8a$$
$$-4.9 = -T_B + 0.5a$$
$$T_A = \frac{1372}{75} = 18.3 \text{ N}$$

$T_A = 18$ N (2 SF – AQA)

$$T_B = \frac{392}{75} = 5.23 \text{ N}$$

$T_B = 5.2$ N (2 SF – AQA)

7 b)

$$a = \frac{49}{75} = 0.653 \text{ ms}^{-2}$$

$a = 0.65$ ms^{-2} (2 SF – AQA)

A is accelerating downwards

B is accelerating to the left

C is accelerating upwards

7 c)

The string is light, inextensible and taut. The boxes are modelled as particles.

8)

On A:

$$\mathrm{R}(\uparrow): \quad R - Mg = 0 \Rightarrow R = Mg$$
$$F = \mu Mg$$
$$\mathrm{R}(\rightarrow): \quad T - \mu Mg = Ma$$

On B:

$$\mathrm{R}(\downarrow): \quad mg - T = ma$$
$$mg - \mu Mg = (M + m)a \Rightarrow a = \frac{mg - \mu Mg}{M + m} \text{ ms}^{-2}$$
$$\frac{mg - \mu Mg}{M + m} \geq 0 \Rightarrow mg - \mu Mg \geq 0 \Rightarrow mg \geq \mu Mg$$
$$\therefore m \geq \mu M \text{ as req.}$$